

401. Binary Watch

Solved 

Easy

 Topics

 Companies

 Hint

A binary watch has 4 LEDs on the top to represent the hours (0-11), and 6 LEDs on the bottom to represent the minutes (0-59). Each LED represents a zero or one, with the least significant bit on the right.

- For example, the below binary watch reads "4:51".



Given an integer `turnedOn` which represents the number of LEDs that are currently on (ignoring the PM), return *all possible times the watch could represent*. You may return the answer in **any order**.

The hour must not contain a leading zero.

- For example, "01:00" is not valid. It should be "1:00".

The minute must consist of two digits and may contain a leading zero.

- For example, "10:2" is not valid. It should be "10:02".

Example 1:

Input: turnedOn = 1

Output:

["0:01", "0:02", "0:04", "0:08", "0:16", "0:32", "1:00", "2:00", "4:00", "8:00"]

Example 2:

Input: turnedOn = 9

Output: []

Constraints:

- $0 \leq \text{turnedOn} \leq 10$

Python:

class Solution:

def readBinaryWatch(self, k: int) -> List[str]:

if k == 0: return ["0:00"]

if k > 8: return []

res = []

q = (1 << k) - 1

while q < (1 << 10):

time = self.isValid(q)

if time:

res.append(time)

r = q & -q

n = q + r

q = (((n ^ q) >> 2) // r) | n

return res

```
def isValid(self, q: int) -> str:
    hour = q >> 6
    min = q & 63

    if hour >= 12 or min >= 60:
        return ""

    return f"{hour}:{min:02d}"
```

JavaScript:

```
const readBinaryWatch = k => {
    if (k === 0) return ["0:00"];
    if (k > 8) return [];
    const res = [];

    let q = (1 << k) - 1;

    while (q < (1 << 10)) {
        const time = isValid(q);
        if (time) res.push(time);

        const r = q & -q;
        const n = q + r;
        q = (((n ^ q) >> 2) / r) | n;
    }

    return res;
};

const isValid = q => {
    const hour = q >> 6;
    const min = q & 63;

    if (hour >= 12 || min >= 60) return "";

    return `${hour}:${String(min).replace(/^(\\d)$/, "0$1")}`;
};
```

Java:

```
class Solution {
    public List<String> readBinaryWatch(int k) {
        if (k == 0) return Arrays.asList("0:00");
        if (k > 8) return new ArrayList<>();
```

```

List<String> res = new ArrayList<>();
int q = (1 << k) - 1;

while (q < (1 << 10)) {
    String time = isValid(q);
    if (!time.isEmpty())
        res.add(time);

    int r = q & -q;
    int n = q + r;
    q = (((n ^ q) >> 2) / r) | n;
}

return res;
}

String isValid(int q) {
    int hour = q >> 6;
    int min = q & 63;
    if (hour >= 12 || min >= 60) return "";
    return String.format("%d:%02d", hour, min);
}
}

```